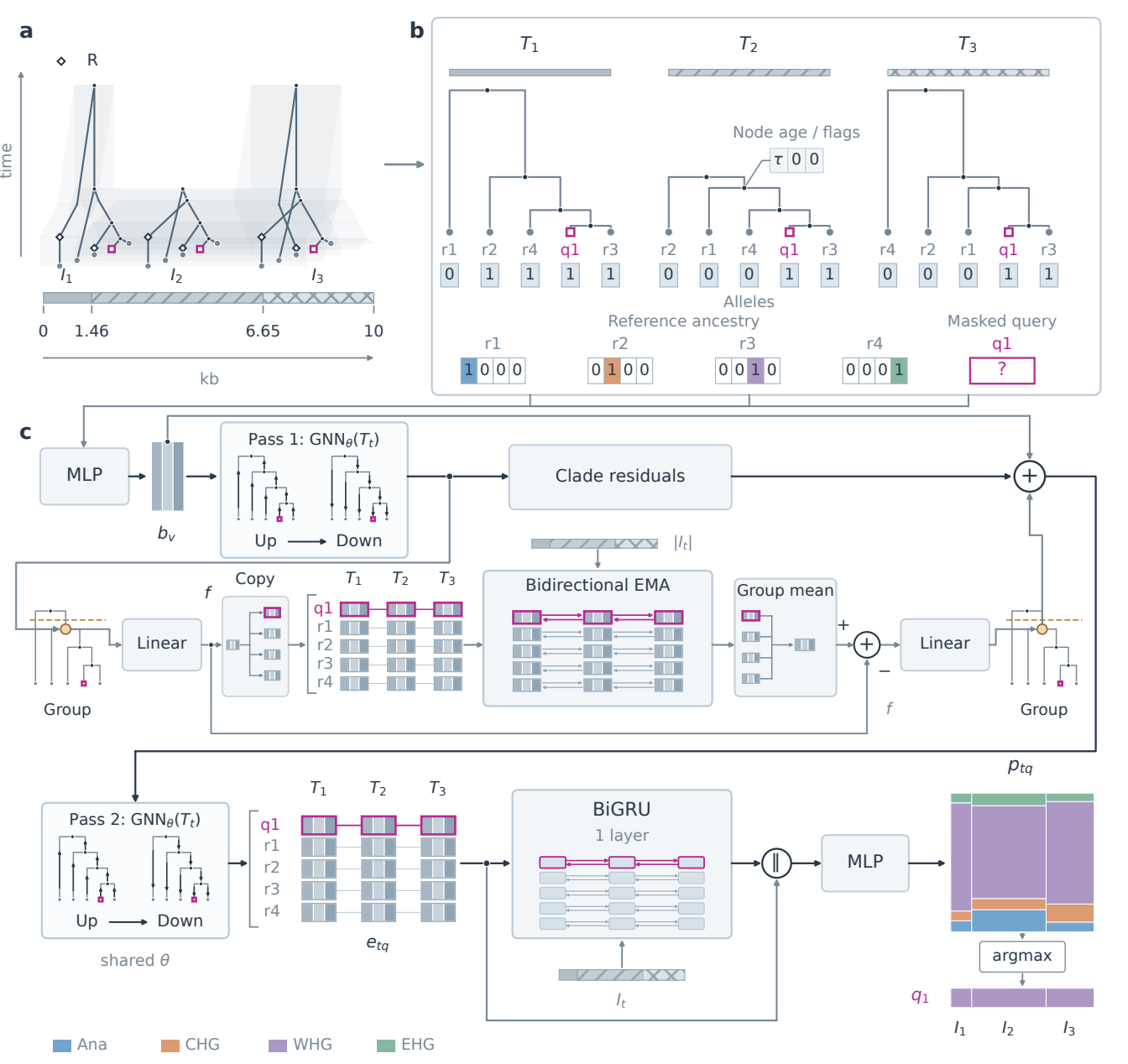

a
R
time
I_1
I_2
I_3
0
1.46
6.65
10
kb
b
T_1
T_2
T_3
Node age / flags
τ 0 0
r1 r2 r4 q1 r3
0 1 1 1 1
r2 r1 r4 q1 r3
0 0 0 1 1
r4 r2 r1 q1 r3
0 0 0 1 1
Alleles
Reference ancestry
Masked query
r1
1 0 0 0
r2
0 1 0 0
r3
0 0 1 0
r4
0 0 0 1
q1
?
c
MLP
b_v
Pass 1: GNN$_\theta(T_t)$
Up → Down
Clade residuals
$|I_t|$
Bidirectional EMA
Group mean
Copy
f
Linear
Group
T_1 T_2 T_3
q1
r1
r2
r3
r4
+
−
Linear
Group
f
p_{tq}
Pass 2: GNN$_\theta(T_t)$
Up → Down
shared θ
e_{tq}
BiGRU
1 layer
I_t
MLP
argmax
q_1
I_1 I_2 I_3
Ana
CHG
WHG
EHG